GROUP NUMBER: 23

GROUP MEMBERS: **Angelica Zonta (2158814) Hermann Serain (2158294)**

TEST: The goal is to test the performance of **Sticky Sampling** (**SS**) and **Count-Min** (**CM**) for different input configurations. We refer to an item as *frequent/almost frequent/rare* if its true frequency is in the interval ¿ / $[n(\phi-\epsilon), n\phi)$, ¿ respectively.

You must fill in the following tables using the input configurations indicated in red for each table. You must run each test three times and report the average values of the three runs

STICKY SAMPLING: n=1000000, ϕ=0.07, δ=0.05, port=8888				
ϵ	**Number of frequent items returned by SS**	**Number of almost frequent items returned by SS**	**Number of rare items returned by SS**	**Number of elements stored in the dictionary used by SS**
0.01	10	0	0	37
0.02	10	0	0	23
0.04	10	2	0	18

COUNT MIN: n=1000000, ϕ=0.07, d=5, port=8888				
w	**Number of frequent items returned by CM**	**Number of almost frequent items (w.r.t. ϵ=0.04) returned by CM**	**Number of rare items (w.r.t. ϵ=0.04) returned by CM**	**Total number of items returned by CM (sum of previous 3 columns)**
15	10	0	196	206
30	10	0	10	20
60	10	0	2	12

ANSWER THE FOLLOWING QUESTIONS

- Did you use some Generative AI tool? YES
- (Only if you answered YES to previous question) Describe in at most 10 lines how you used the Generative AI tool.
  As in the previous homework, artificial intelligence was used only during an initial and limited exploratory phase of the project. The first two functions (SS and CM algorithms) were fully designed and implemented independently by the two group members without any use of AI tools. To perform the 3 runs we have used AI to adapt the code on saving the 3 results for each run and then compute the average. Finally we have pushed the original code consisting of just one run as requested from the submission.